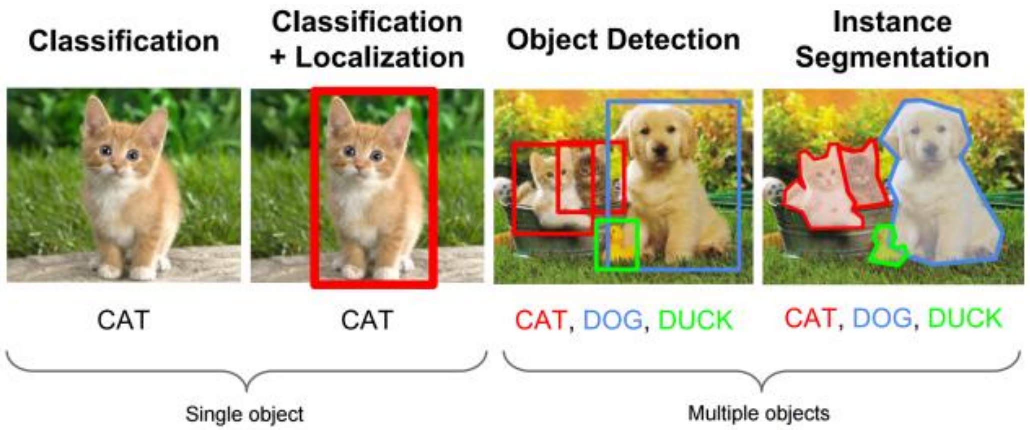
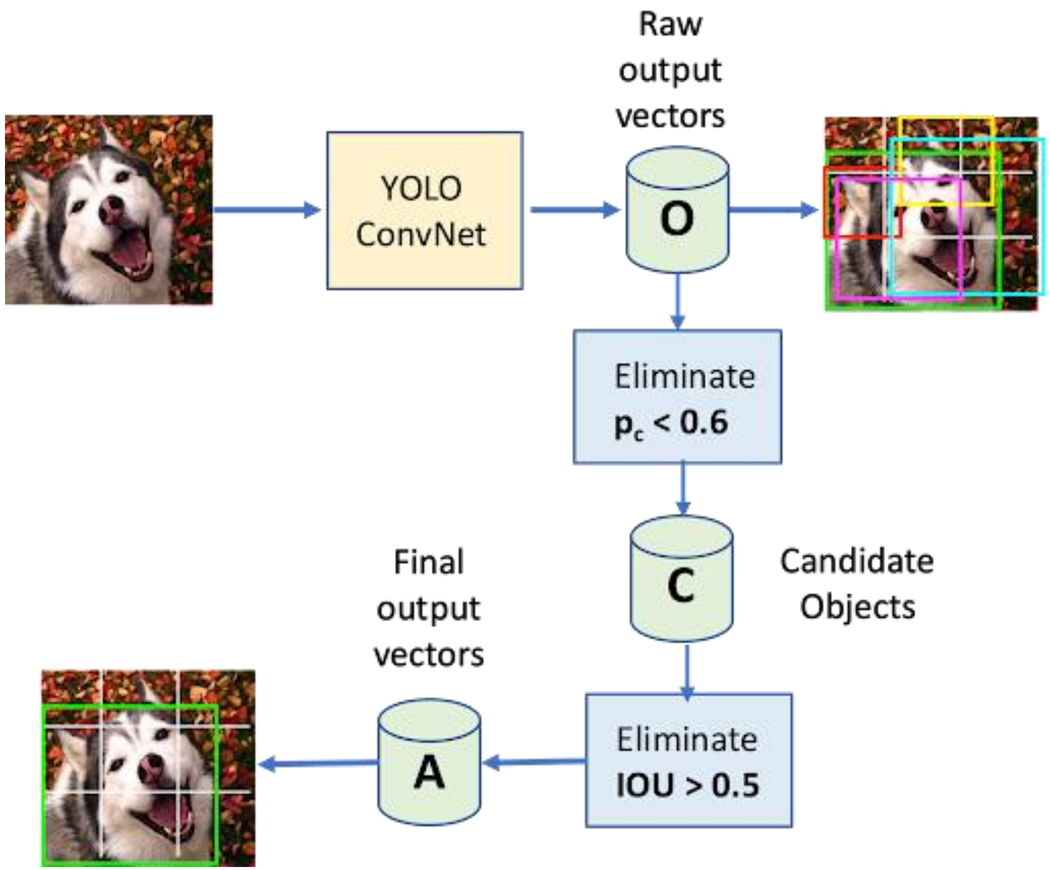
Localization: Predicting Where the Objects Are



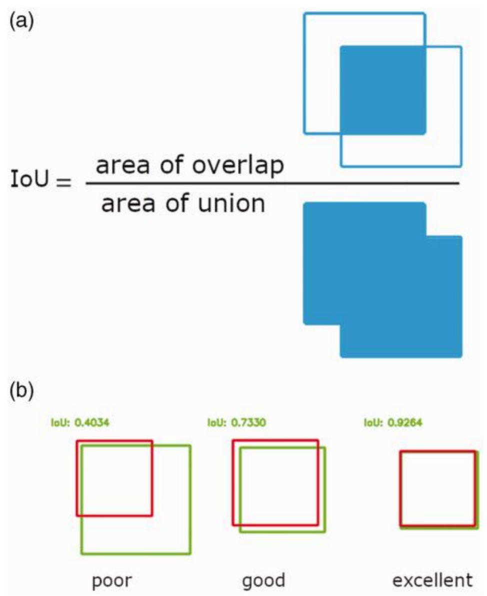
- Unlike classification (which says "what" is in the image), localization identifies "where" the object is.
- We represent the location of an object using a **Bounding Box**.
- **Coordinates**: Standard bounding boxes use a 4-value vector: [x, y, w, h].
- **x, y**: The center (or top-left) coordinates of the box.
- **w, h**: The width and height of the box relative to the image size.

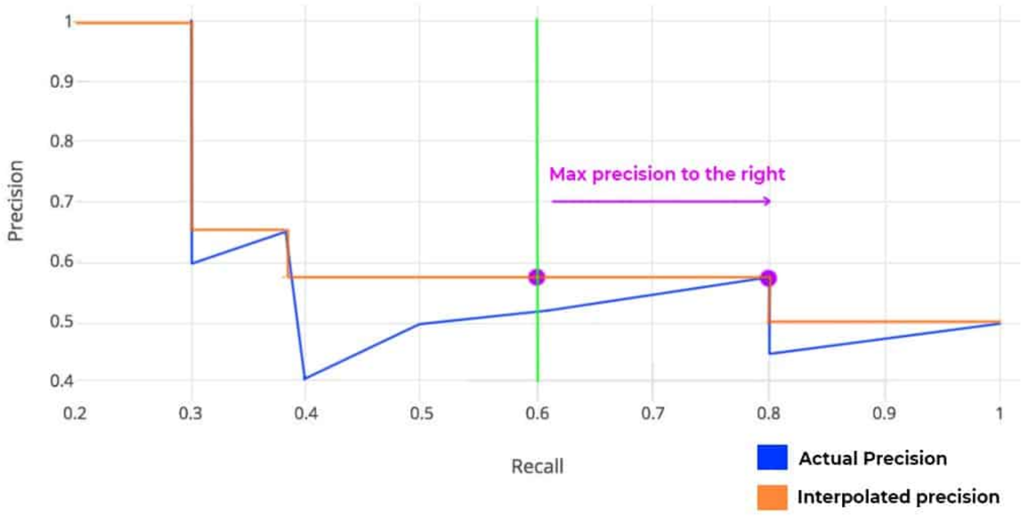


YOLO: Real-Time Detection Logic

- YOLO is a single-stage detector that views the entire image once to predict both boxes and classes.
- It is designed for high-speed, real-time performance on video streams.
- **Evolution**: The architecture has evolved from YOLOv8 to the latest YOLOv11 for improved accuracy and efficiency.
- The model divides the image into a grid and predicts probabilities for each grid cell.

Evaluating Performance: IoU and mAP





- **Intersection over Union (IoU)**: Measures the overlap between the predicted box and the actual "ground truth" box.
- **Formula**:$IoU = \frac{\text{Area of Overlap}}{\text{Area of Union}}$. A score of 1.0 is a perfect match.
- **Mean Average Precision (mAP)**: The primary metric for object detection that summarizes precision and recall across all categories.